CS2201 Computer Organization & Architecture

L24: Advanced Cache Optimization Techniques-1



John Jose

Associate Professor

Department of Computer Science & Engineering

Indian Institute of Technology Guwahati

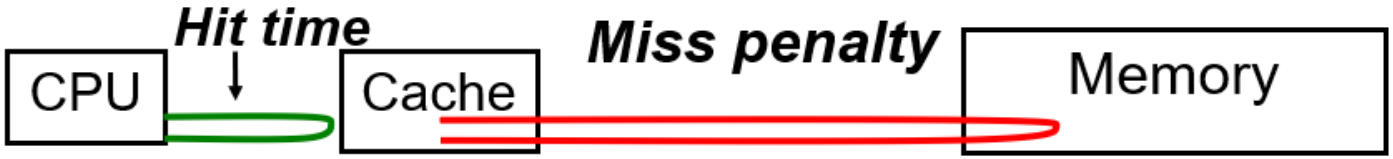
Accessing Cache Memory



Average memory access time = Hit time + (Miss rate×Miss penalty)

- **Hit Time:** Time to find the block in the cache and return it to processor *[indexing, tag comparison, transfer].*
- **Miss Rate:** Fraction of cache access that result in a miss.
- **Miss Penalty:** Number of additional cycles required upon encountering a miss to fetch a block from the next level of memory hierarchy.

How to optimize cache ?



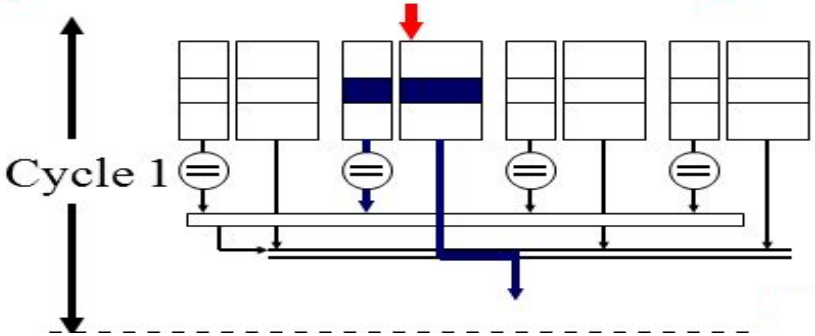
Average memory access time = Hit time + (Miss rate×Miss penalty)

- Motives
  - Reducing the miss rate
  - Reducing the miss penalty
  - Reducing the hit time

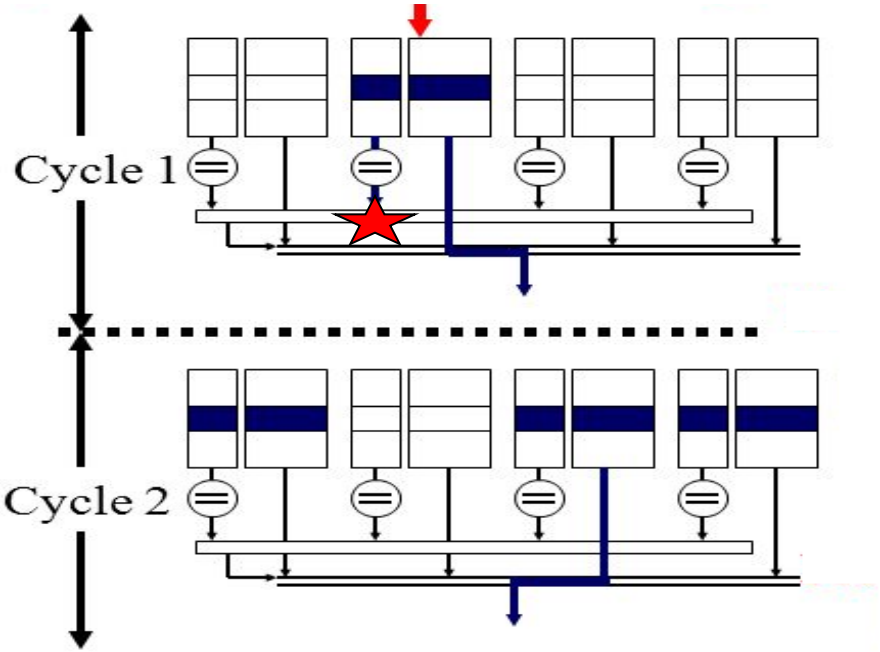
Way Prediction

- **Predict the way in a set to reduce hit time**
- To improve hit time, predict the way to pre-set multiplexer
- Extra bits are set to predict the block with in the set
- Remember the MRU way of a given set
- Using the prediction bits, power gating can be done on unused ways for reducing power.

Way Prediction



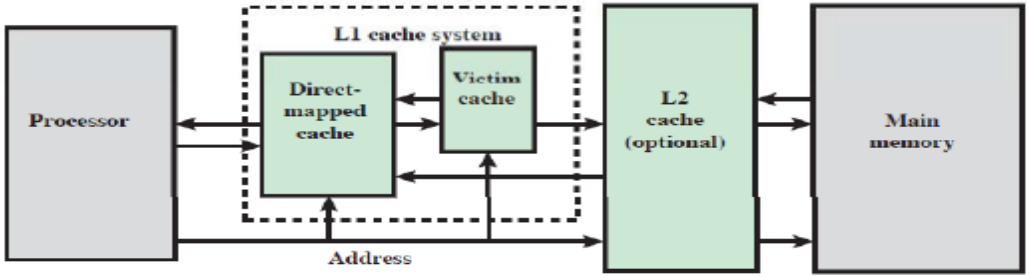
Way Prediction - Correct



Way Prediction Miss

Victim Caches

- **Introduce victim caches to reduce mis- penalty**
- Additional storage near L1 for MR evicted blocks
- Efficient for thrashing problem in L1 cache
- Look up Victim Caches before L2
- L1 and Victim caches are exclusive



Pipelining Cache

- **Pipelined cache for faster clock cycle time.**
- Split cache memory access into several sub-stages
- Ex: Indexing, Tag Read, Hit/Miss Check, Data Transfer
- Pipeline cache access to improve bandwidth
  - Examples: Pentium 4 – Core i7: 4 cycles
- But slow in hits.
- Increases branch mis-prediction penalty
- Makes it easier to increase associativity



johnjose@iitg.ac.in

http://www.iitg.ac.in/johnjose/

https://www.linkedin.com/in/john-jose-bb8431106/